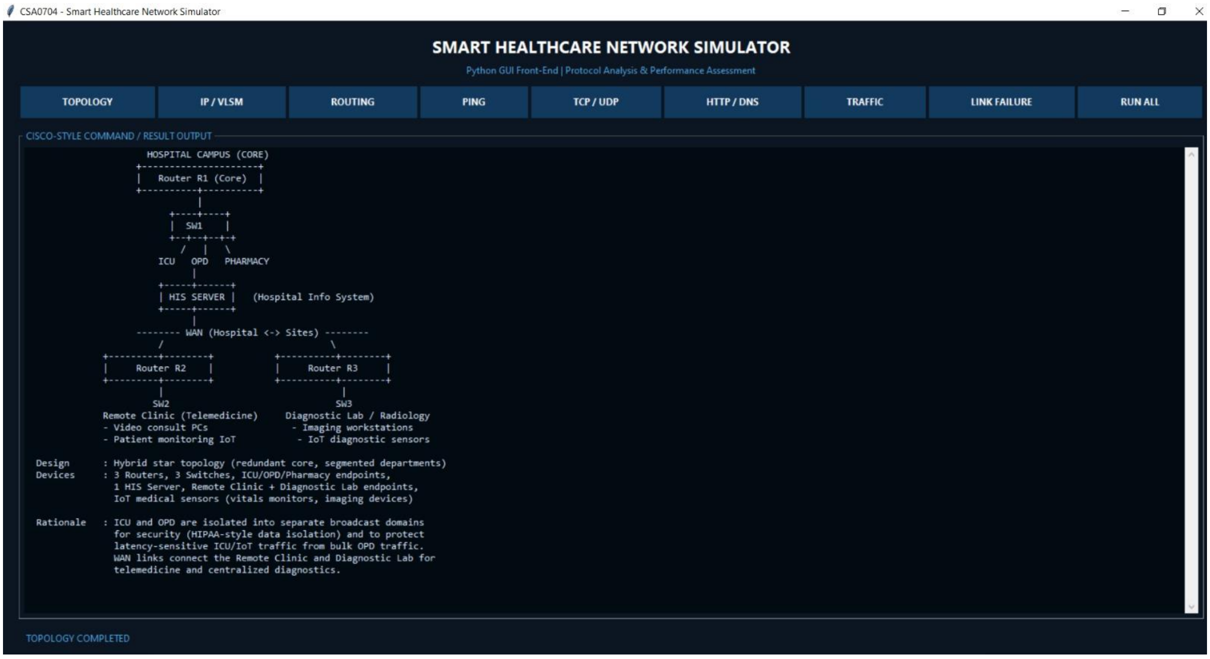
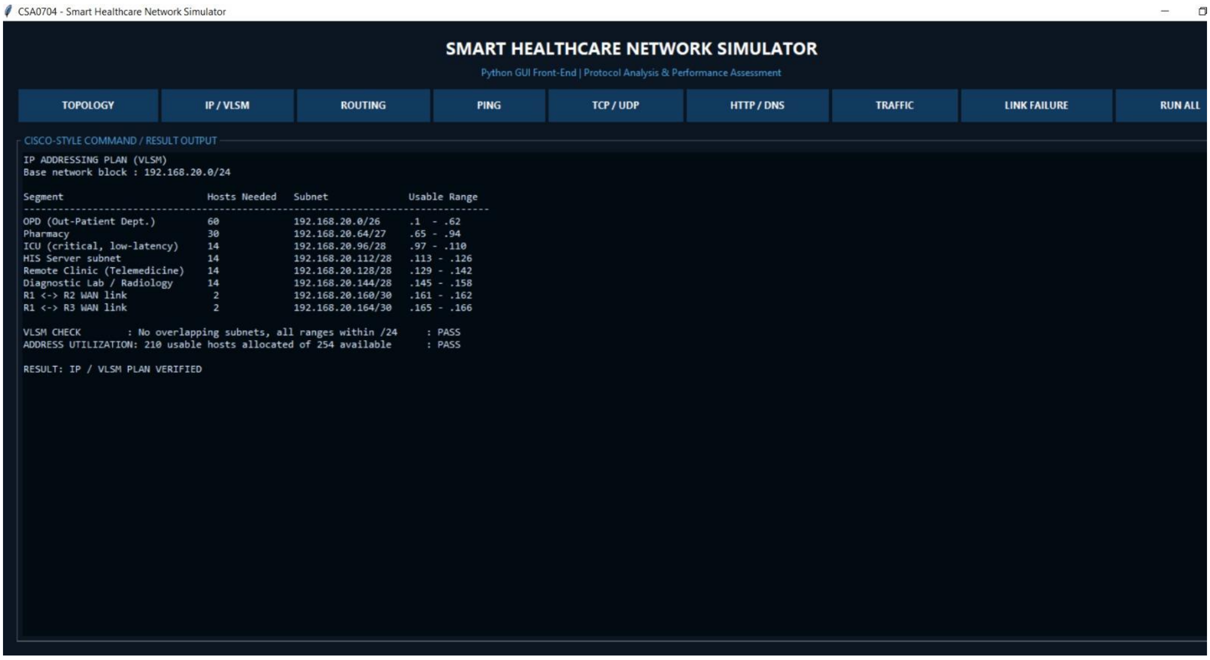
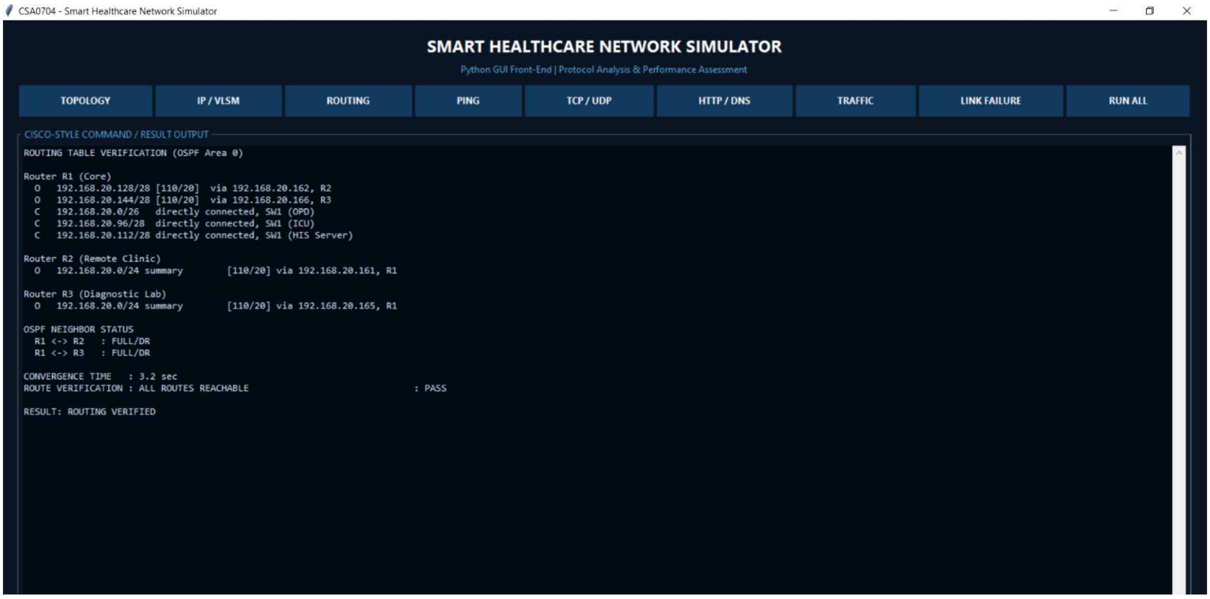
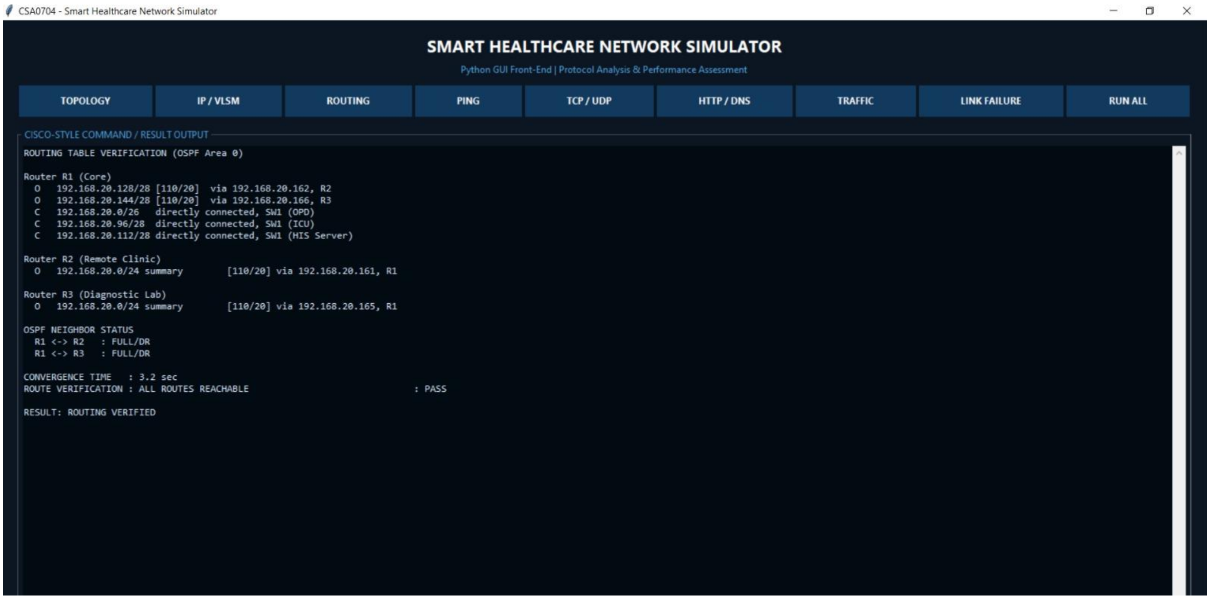
IMPLEMENTATION :






CSA0704 - Smart Healthcare Network Simulator
SMART HEALTHCARE NETWORK SIMULATOR
Python GUI Front-End | Protocol Analysis & Performance Assessment
TOPOLOGY
IP / VLSM
ROUTING
PING
TCP / UDP
HTTP / DNS
TRAFFIC
LINK FAILURE
RUN ALL
CISCO-STYLE COMMAND / RESULT OUTPUT
ROUTING TABLE VERIFICATION (OSPF Area 0)
Router R1 (Core)
O 192.168.20.128/28 [110/20] via 192.168.20.162, R2
O 192.168.20.144/28 [110/20] via 192.168.20.166, R3
C 192.168.20.0/26 directly connected, SW1 (OPD)
C 192.168.20.96/28 directly connected, SW1 (ICU)
C 192.168.20.112/28 directly connected, SW1 (HIS Server)
Router R2 (Remote Clinic)
O 192.168.20.0/24 summary [110/20] via 192.168.20.161, R1
Router R3 (Diagnostic Lab)
O 192.168.20.0/24 summary [110/20] via 192.168.20.165, R1
OSPF NEIGHBOR STATUS
R1 <-> R2 : FULL/DR
R1 <-> R3 : FULL/DR
CONVERGENCE TIME : 3.2 sec
ROUTE VERIFICATION : ALL ROUTES REACHABLE : PASS
RESULT: ROUTING VERIFIED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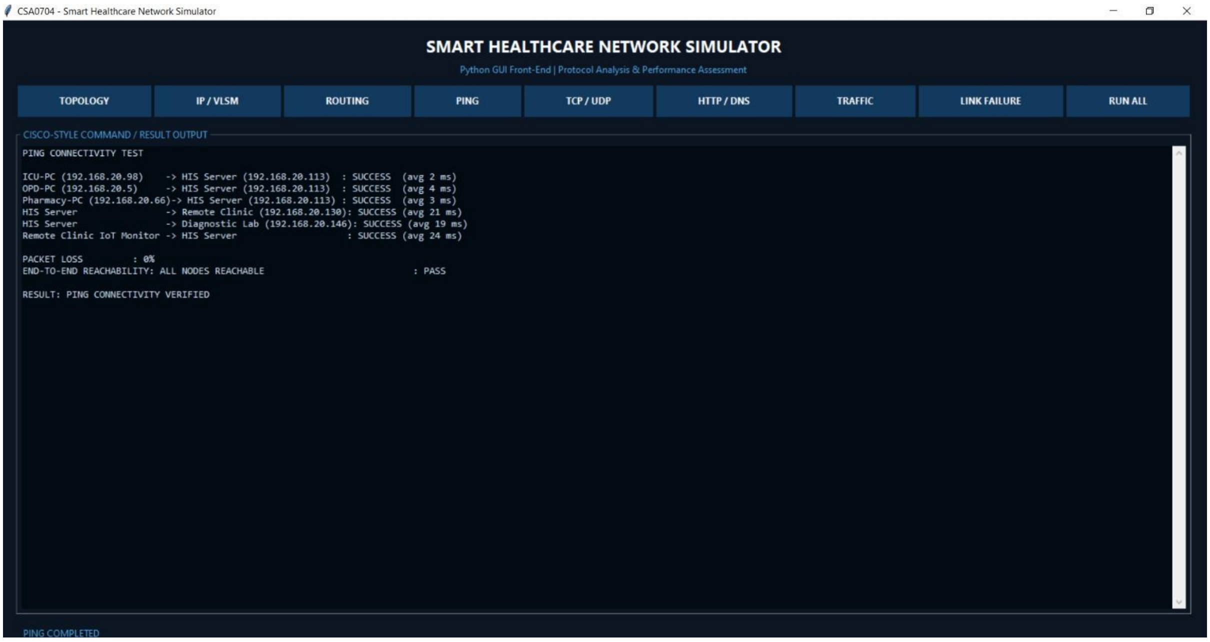
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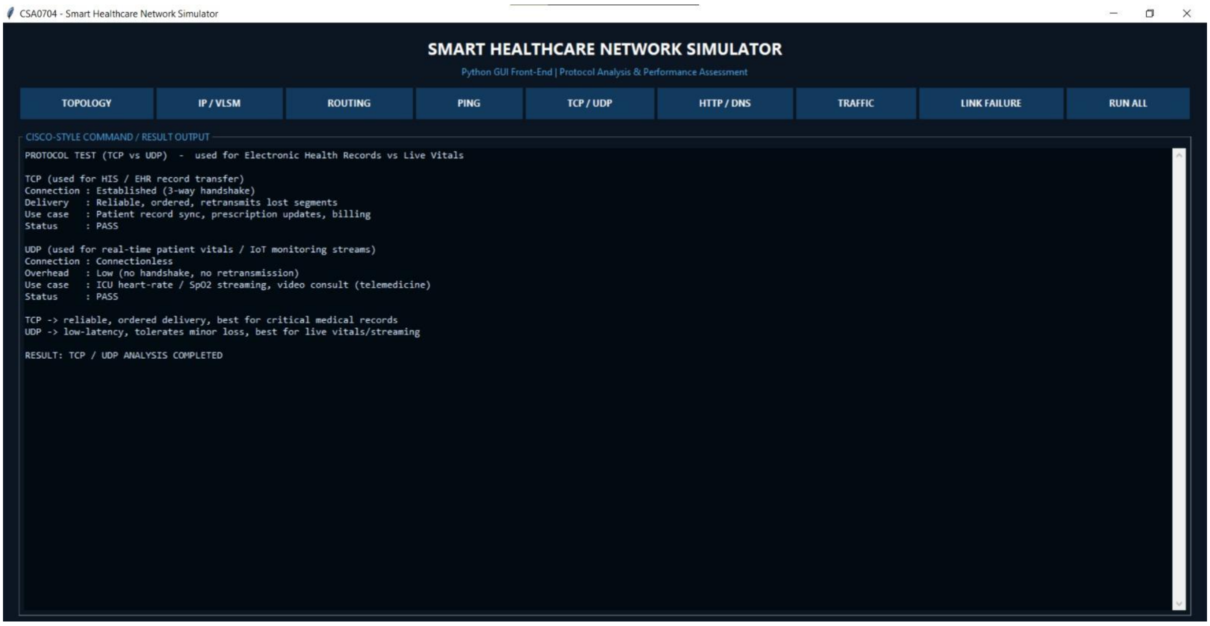
PING CONNECTIVITY TEST
ICU-PC (192.168.20.98)   -> HIS Server (192.168.20.113) : SUCCESS (avg 2 ms)
OPD-PC (192.168.20.5)   -> HIS Server (192.168.20.113) : SUCCESS (avg 4 ms)
Pharmacy-PC (192.168.20.66)-> HIS Server (192.168.20.113) : SUCCESS (avg 3 ms)
HIS Server   -> Remote Clinic (192.168.20.130): SUCCESS (avg 21 ms)
HIS Server   -> Diagnostic Lab (192.168.20.146): SUCCESS (avg 19 ms)
Remote Clinic IoT Monitor -> HIS Server   : SUCCESS (avg 24 ms)
PACKET LOSS   : 0%
END-TO-END REACHABILITY: ALL NODES REACHABLE   : PASS
RESULT: PING CONNECTIVITY VERIFIED
PING COMPLETED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PROTOCOL TEST (TCP vs UDP) - used for Electronic Health Records vs Live Vitals
TCP (used for HIS / EHR record transfer)
Connection : Established (3-way handshake)
Delivery   : Reliable, ordered, retransmits lost segments
Use case   : Patient record sync, prescription updates, billing
Status   : PASS
UDP (used for real-time patient vitals / IoT monitoring streams)
Connection : Connectionless
Overhead   : Low (no handshake, no retransmission)
Use case   : ICU heart-rate / SpO2 streaming, video consult (telemedicine)
Status   : PASS
TCP -> reliable, ordered delivery, best for critical medical records
UDP -> low-latency, tolerates minor loss, best for live vitals/streaming
RESULT: TCP / UDP ANALYSIS COMPLETED


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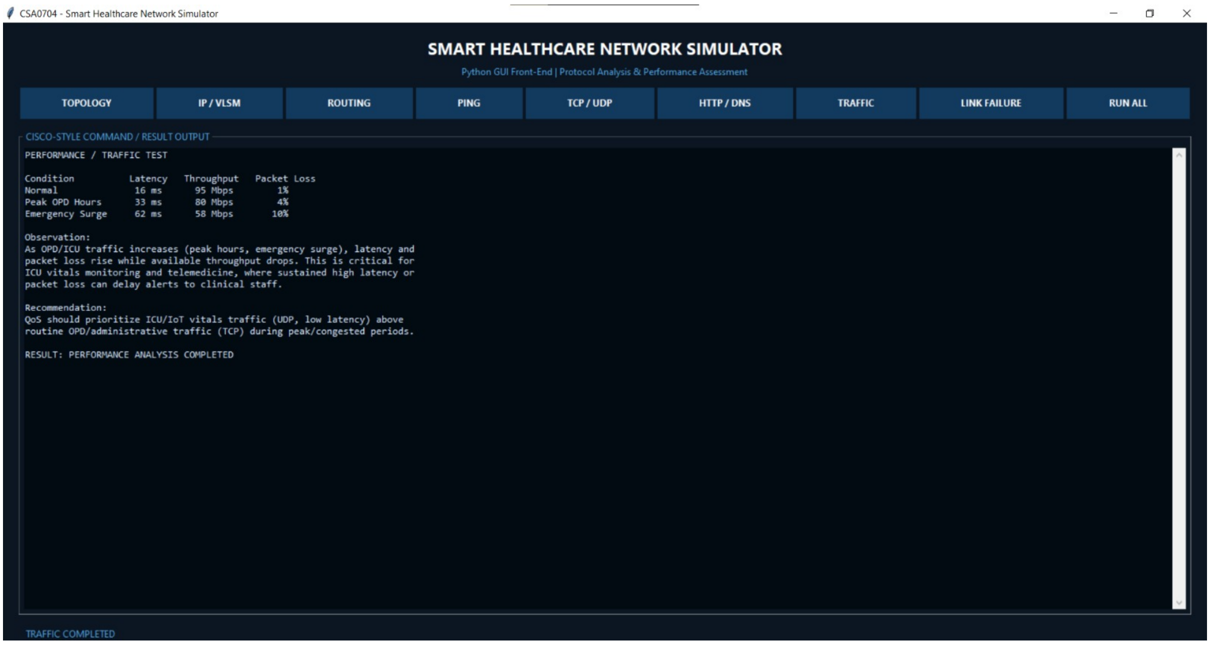
CISCO-STYLE COMMAND / RESULT OUTPUT
HIS SERVER
IP Address : 192.168.20.113
HTTP SERVICE (Hospital Patient Portal)
Status : ON
Request : http://his.hospital.local
Result : Web response received (Patient Portal loaded)
DNS SERVICE
Status : ON
his.hospital.local -> 192.168.20.113
clinic.hospital.local -> 192.168.20.130
lab.hospital.local -> 192.168.20.146
Result : Name resolution successful for all hospital domains
RESULT: HTTP / DNS VERIFIED
HTTP / DNS COMPLETED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PERFORMANCE / TRAFFIC TEST
Condition Latency Throughput Packet Loss
Normal 16 ms 95 Mbps 1%
Peak OPD Hours 33 ms 80 Mbps 4%
Emergency Surge 62 ms 58 Mbps 10%
Observation:
As OPD/ICU traffic increases (peak hours, emergency surge), latency and packet loss rise while available throughput drops. This is critical for ICU vitals monitoring and telemedicine, where sustained high latency or packet loss can delay alerts to clinical staff.
Recommendation:
QoS should prioritize ICU/IoT vitals traffic (UDP, low latency) above routine OPD/administrative traffic (TCP) during peak/congested periods.
RESULT: PERFORMANCE ANALYSIS COMPLETED
TRAFFIC COMPLETED


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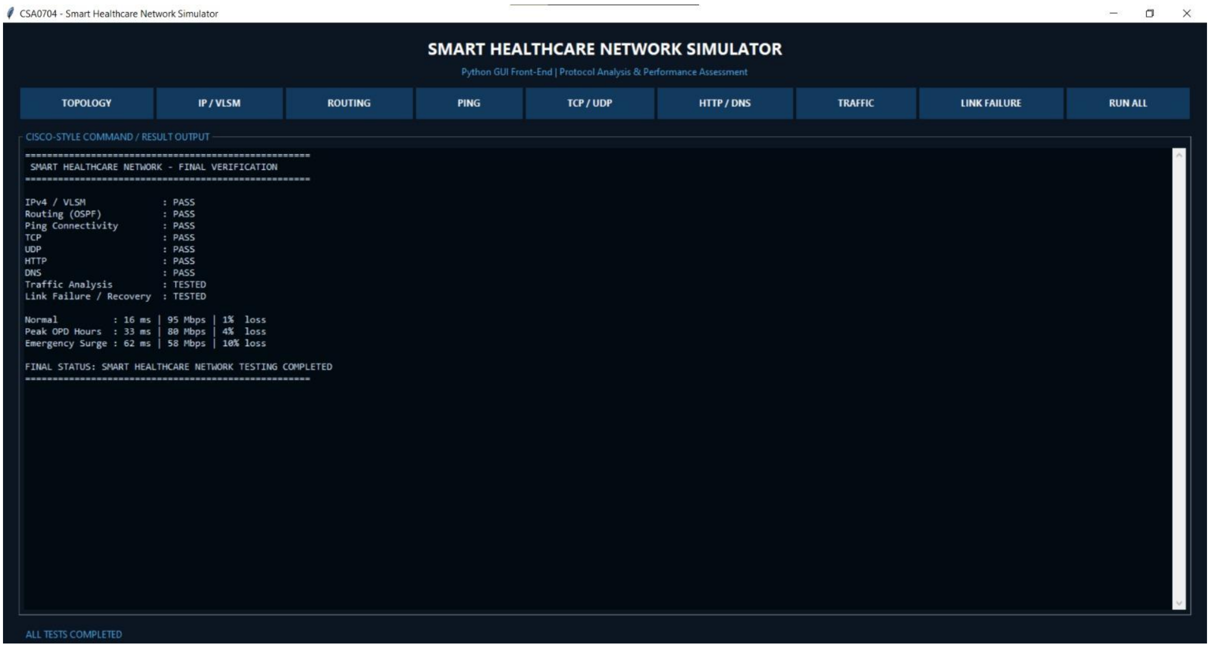
LINK FAILURE TEST (Critical Path: ICU Switch <-> Core Router)
BEFORE FAILURE
ICU-PC -> HIS Server : SUCCESS
LINK DOWN
R1 G0/1 -------- X -------- SW1 (ICU)
DURING FAILURE
ICU-PC -> HIS Server : REQUEST TIMED OUT
ALERT : ICU vitals monitoring stream INTERRUPTED
RECOVERY
Backup path activated via redundant OSPF route
Link restored
ICU-PC -> HIS Server : SUCCESS
IMPACT NOTE:
In a real hospital deployment, this link should have hardware redundancy
(dual NICs / secondary switch uplink) since ICU connectivity loss directly
affects patient safety monitoring.
RESULT: FAILURE AND RECOVERY VERIFIED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SMART HEALTHCARE NETWORK - FINAL VERIFICATION
IPv4 / VLSM : PASS
Routing (OSPF) : PASS
Ping Connectivity : PASS
TCP : PASS
UDP : PASS
HTTP : PASS
DNS : PASS
Traffic Analysis : TESTED
Link Failure / Recovery : TESTED
Normal : 16 ms | 95 Mbps | 1% loss
Peak OPD Hours : 33 ms | 80 Mbps | 4% loss
Emergency Surge : 62 ms | 58 Mbps | 10% loss
FINAL STATUS: SMART HEALTHCARE NETWORK TESTING COMPLETED
ALL TESTS COMPLETED